DETAILED DESCRIPTION OF THE INVENTION

Method of Generating a User-Specific Penile Anatomical Geometry Profile

In various embodiments, the disclosure further provides a method of generating a user-specific penile anatomical geometry profile comprising the steps of: initializing a scanning device including at least one time-of-flight (ToF) ranging sensor and at least one motion sensing component; establishing a defined anatomical starting reference position; and verifying operational readiness through calibration or signal validation routines.

The method may further comprise emitting a ranging signal from the ToF sensor toward a penile anatomical surface; detecting reflected return signals; calculating radial distance measurements relative to a defined reference plane; and concurrently detecting longitudinal displacement of the device relative to the anatomical surface using one or more optical motion sensors, inertial sensors, or hybrid displacement detection systems.

The method may include synchronizing distance measurements with positional displacement data to generate a temporally indexed dataset representing radial surface values along a longitudinal axis. The method may further comprise filtering distance values to remove noise or outliers; compensating for angular deviation or motion artifacts; and interpolating intermediate surface points between sampled positions.

In certain embodiments, the method includes constructing cross-sectional profiles at discrete longitudinal intervals; computing curvature vectors, taper gradients, symmetry indices, and volumetric estimations; and generating a reconstructed three-dimensional geometric representation of the penile anatomy.

The method may further comprise encoding the reconstructed geometry into a standardized digital geometry profile; generating derived structural metrics; and optionally normalizing such metrics against stored reference distributions to produce comparative indices or classification outputs.

In some implementations, the method includes validating measurement authenticity using physiological verification signals; encrypting collected data prior to storage or transmission; and securely transmitting the encoded geometry profile to a companion application or remote processing environment.

The method steps described herein may be executed sequentially, partially in parallel, or adaptively reordered depending on device configuration and processing architecture. Variations in sensing modality, sampling density, reconstruction technique, or encoding format do not depart from the inventive concept of generating a user-specific penile anatomical geometry profile through synchronized radial distance acquisition and motion-based positional indexing.